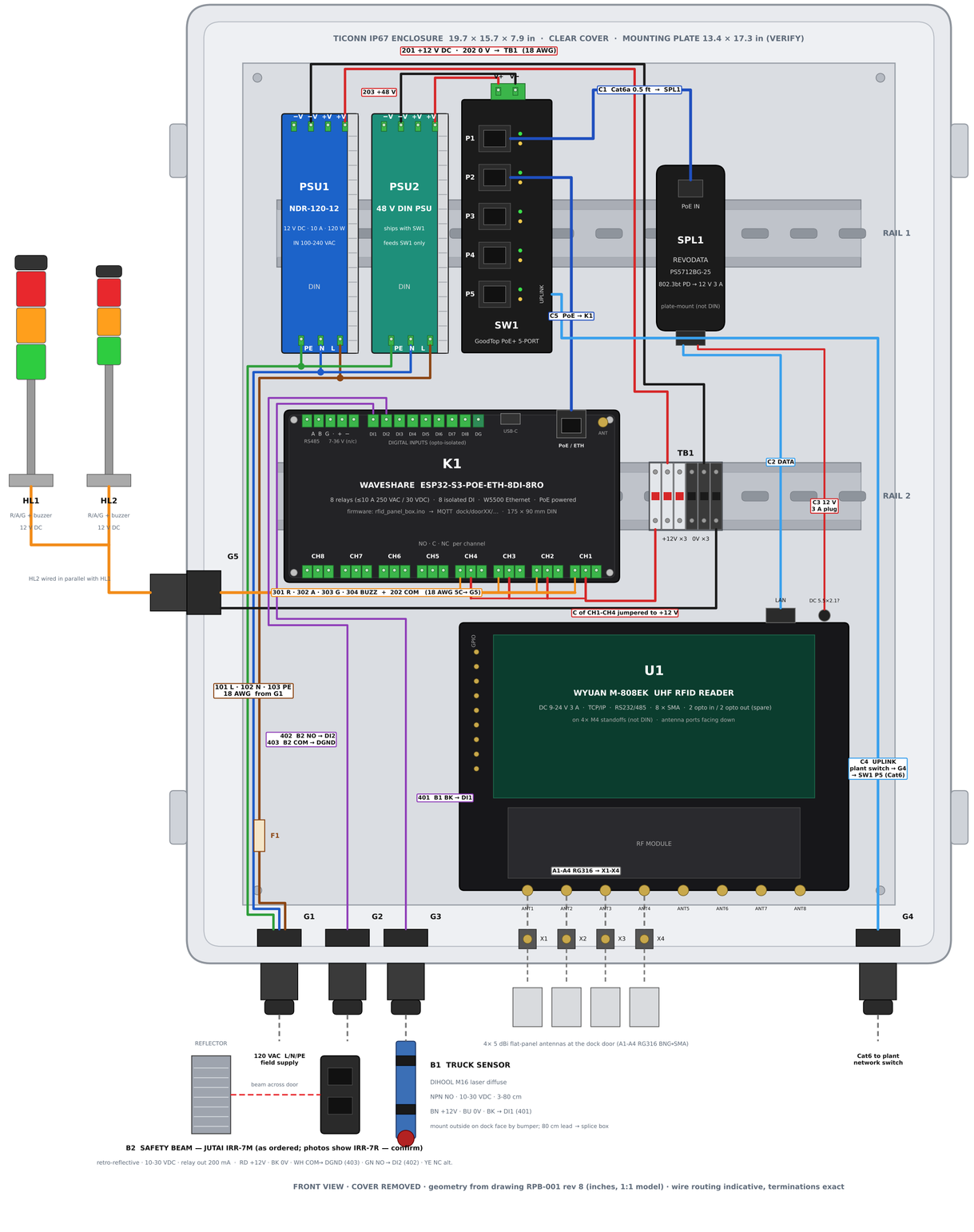
RFID PANEL BOX · RPB-001 · REV 8

Gielow Pickles · G5 Croswell · Inventory Control / Automation · Illustrated schematic, Saturday Oct 17, 2026 · matches drawing sheets 1-2 (rev 8) and parts list rev 5



FRONT VIEW · COVER REMOVED · geometry from drawing RPB-001 rev 8 (inches, 1:1 model) · wire routing indicative, terminations exact

WIRE & CABLE LEGEND

L (brown, 101)	N (blue, 102)
PE (gn/ye, 103)	+12 V / +48 V DC (red)
0 V DC (black, 202)	PoE data + power (blue)
Ethernet data only (lt blue)	sensor signal (purple)
stack-light cable (orange)	antenna coax / field (dashed)

POWER TREE

120 VAC	G1 → F1 → PSU1 & PSU2 (parallel)
PSU1 12 V	NDR-120-12 → TB1 → relay commons, HL1/HL2, B1, B2 (~11 W of 120 W)
PSU2 48 V	ships with SW1 → SW1 V+/V−
SW1 PoE+	P1 → SPL1 (reader) · P2 → K1 (board) 4×30 W ports — verify total budget
SPL1 out	12 V 3 A plug + data pigtail → U1 (≤25.5 W from a 30 W port: test at full RF)
K1	PoE only — leave its 7-36 V terminal empty
Isolation	U1 DC− rides on SPL1 (not bonded). K1 DGND bonded to 0 V (202) — verify vs Waveshare DI diagram

K1 RELAY & INPUT MAP

CH1	RED	no truck at dock
CH2	AMBER	truck docked, reader idle
CH3	GREEN	beam broken → reader active
CH4	BUZZER	alert (optional)
CH5-8	spare	—
DI1	B1 truck	1 = truck present (NPN NO)
DI2	B2 beam	1 = beam broken (NO/NC on bench)
DGND	B2 COM	bonded to 0 V
DI3-8	spare	—

Logic lives in the C# service; the box only reports inputs and sets the light.
Fail-safe: red on power-up and after 10 s without the server.

WIRE / CABLE LIST (terminations per sheet 2)

No.	Signal	From → To	Color / type
101-103	AC L / N / PE	G1 → F1 → PSU1 L/N/PE, jumpered to PSU2	BN / BU / GNYE 18 AWG
201 / 202	+12 V / 0 V	PSU1 +V/−V → TB1; TB1 → K1 C1-C4, HL COM, B1, B2	RD / BK 18 AWG
203	+48 V	PSU2 +V/−V → SW1 V+/V−	RD / BK 18 AWG
301-304	stack light	K1 CH1-CH4 NO → HL1 (HL2 parallel) + 202 COM	18 AWG 5-cond via G5
401	B1 truck	DIHOOL BK → K1 DI1 (BN +12 V, BU 0 V from TB1)	22 AWG via G3
402 / 403	B2 beam	JUTAI GN (NO) → DI2; WH (COM) → DGND; RD/BK supply	22 AWG via G2
C1 / C5	PoE	SW1 P1 → SPL1; SW1 P2 → K1	Cat6a 0.5 ft ×2
C2 / C3	reader	SPL1 data pigtail → U1 LAN; SPL1 12 V plug → U1 DC	splitter pigtails
C4	uplink	plant network → G4 → SW1 P5	Cat6 (not on list)
A1-A4	antenna	U1 ANT1-4 → X1-X4 bulkheads → 4 panel antennas	RG316 BNC→SMA

BUILD CHECKS (! verify · ✓ ok)

!	Uplink	C4 (Cat6, plant switch → G4 → SW1 P5) is now on the BOM but unpriced and not yet run. Reader (TCP/IP) and K1 (MQTT) both depend on it.
!	PoE budget	SW1 feeds SPL1 (reader, up to 25.5 W) and K1 (class unknown). Confirm GoodTop total budget; measure K1.
!	Reader power	M-808EK wants 3 A at 12 V (36 W) at full RF. If the splitter browns out, feed U1 from TB1 and use SPL1 for data only.
!	DC plug	SPL1 plug is 5.5×2.5 mm; check U1 jack (often 5.5×2.1). Adapter or land leads on U1 power terminal.
!	K1 inputs	NPN B1 and dry-contact B2 need DGND tied to 0 V. Verify against Waveshare DI wiring; flip INPUT_ACTIVE_LOW in firmware if reversed.
!	B2 model	Alibaba order = IRR-7M; your photos = IRR-7R. Same wiring colors on the sheet you have; confirm on arrival; reflector included?
!	Fit / F1	K1 is 175×90 mm — rail 2 holds K1 + TB1 with 2.9 in spare. F1 (DIN fuse holder, on BOM, unpriced) needs a spot on rail 1 or the spare rail.
✓	Relays	8 relays ≥ 4 needed (both towers in parallel); CH5-8 spare — or drive HL2 separately.
✓	12 V budget	PSU1 ≈ 11 W of 120 W; headroom to power U1 directly if needed.
✓	Fail-safe	Light goes red when the server is unreachable; NO-type truck sensor means a cut wire reads "no truck" (red).